

# Design of All-Digital Phase-Locked Loop

Ming-Wei Hsu

Department of Communications Engineering, National Chung Cheng University

## 1. Introduction

This phase-locked loop consists of a frequency-phase detector, controller, filter, digitally controlled oscillator, and frequency divider, as shown in Figure 1-1. The oscillator and detector are implemented in SPICE; the remaining modules are written in Verilog.

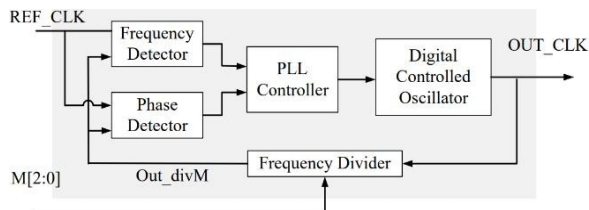


Figure 1-1

## 2. Phase-Frequency Detector

The detector architecture is shown in Figure 2-1. The two inputs are the reference clock and the frequency divider output clock; the two outputs represent whether the divider output clock is leading or lagging. The operating principle is that one output produces a low-pulse while the other remains high, and vice versa. The controller uses the negative edge of the reference clock to detect this change for further processing. UltraSIM simulation results show that the minimum acceptable phase difference between the two inputs is 40 ps, and the maximum input frequency is 1.1 GHz.

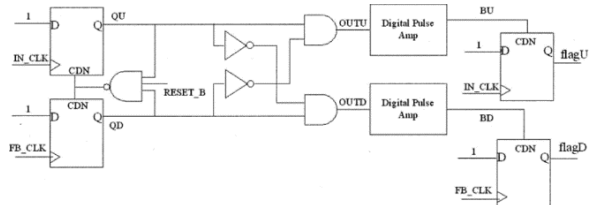


Figure 2-1

## 3. Controller

The controller tracks frequency using a binary search algorithm, as shown in Figure 3-1. Whenever the frequency overshoots the target, the search step is halved. When the step reaches one, frequency acquisition is complete.

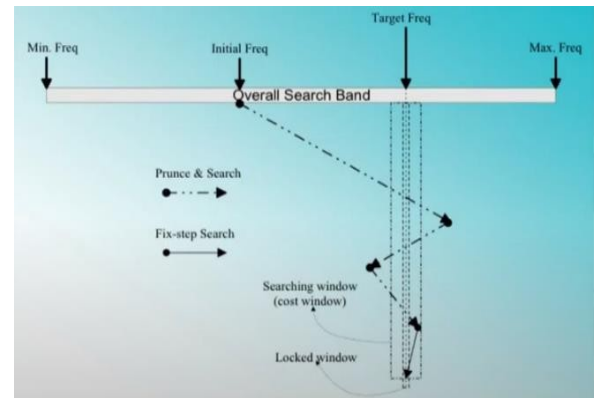


Figure 3-1

## 4. Filter

Once frequency lock is complete, the control code is stored in a register. If a polarity reversal is detected, the control code is reloaded from the register. If the control code increases four consecutive times, the register value increments by one; four consecutive decreases cause it to decrement by one.

## 5. Digitally Controlled Oscillator

The oscillator architecture is shown in Figure 5-1,

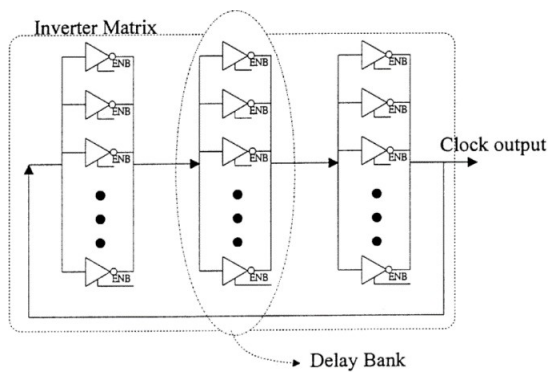


Figure 5-1

consisting of three Delay Banks in series, each with 128 parallel inverters. Because the control code is 7 bits wide and the oscillator input is 128 bits wide — each bit controlling one inverter switch within a Delay Bank — the controller must convert the control code before output. HSPICE simulation results show a maximum output frequency of 2.63 GHz and a minimum of 34 MHz.

## 6. Result Waveforms

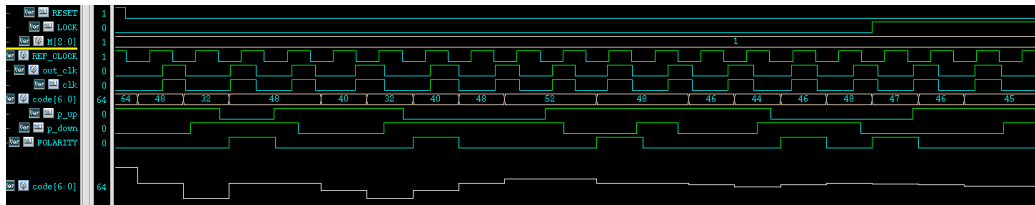


Figure 6-1: Reference clock 1.75 GHz, M=1

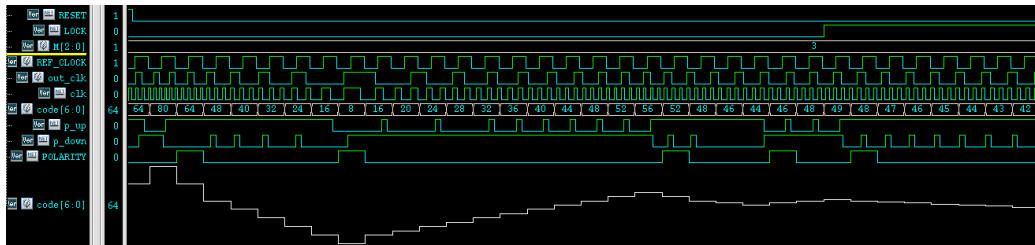


Figure 6-2: Reference clock 400 MHz, M=3

## 7. Specifications

| Para<br>mete<br>rs |                                |        |
|--------------------|--------------------------------|--------|
| 1                  | Target Process                 | 0.18μm |
| 2                  | Max. Reference Clock (MHz)     | 1750   |
| 3                  | Min. Reference Clock (MHz)     | 400    |
| 4                  | Max. Output Clock (MHz)        | 2600   |
| 5                  | Min. Output Clock (MHz)        | 34     |
| 6                  | Max. Divide Ratio              | 7      |
| 7                  | Max. Lock-in Time (# cycle)    | 21     |
| 8                  | Max. Re-lock Time (# cycle)    | 21     |
| 9                  | Max. Output Period Jitter (ps) | 1265   |

|    |                                        |      |
|----|----------------------------------------|------|
| 10 | Max. Output Cycle-to-Cycle Jitter (ps) | 3843 |
| 11 | Max. Output Phase Drift (ps)           | 1984 |
| 12 | Max. Dynamic Power Consumption (mW)    | 39   |
| 13 | Max. Static Power Consumption (mW)     | 1    |

## 8. References

Chung, Ching-Che. (2003). An all-digital phase-locked loop for high-speed clock generation.

Hsu, Terng-Yin. (2001). Design and analysis of a portable high-speed clock generator.

US5,473,285A. (1993). Method and apparatus for performing phase acquisition in an all digital phase lock loop.